

SomaiyaSat Online: Inbound Content Marketing for a PocketQube Mission Website

Course: Digital Marketing (218U04E734) | Lab CA: Poster Presentation
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1 INTRODUCTION

- SomaiyaSat is a 5 cm PocketQube released by SomaiyaPod (use case KJS-SRS-01, with ReOrbit, Finland). After deployment it has one radio, an average power budget under 1 W, and only a short pass over a ground station in low Earth orbit.
- Four downlink modes share that radio: TT&C housekeeping, Codec2 voice, M17 voice and data, and SSTV images. Something onboard has to decide, every second, which frame is both important and actually receivable.
- The audiences — KJSIT and KJSSE students, faculty, the collaborator, and the global amateur-radio (HAM) community — are not in one room. The motivation for this study is to give that mission a public digital channel.
- The real-world product is already deployed at somaiyasat.vercel.app: a mission site plus a ground station whose routing line is computed on the server, not played back as a scripted animation.

2 PROBLEM STATEMENT

- A student space mission is easy to file as a report and hard to understand in a minute. Without a designed journey, a visitor never sees why manual scheduling fails inside a pass, what the AI router does, or that the ground station is actually deciding.
- Scope is the owned website only: Home, Mission, Architecture, Program, and Ground Station. Boundaries: no paid ads and no social campaign in this lab. The site is a demonstration — orbit time, link budget and battery are compressed, and it is not a live spacecraft feed.

3 OBJECTIVES

- Position SomaiyaSat & SomaiyaPod with one offer: a PocketQube that deploys itself, decides what to send home, and serves amateur radio across four multiplexed modes.
- Move a visitor from awareness to action with a primary call to action, “Open ground station”, and a secondary path, “Read the mission”.
- Make the technical proof marketable: every routing choice is scored and explained, so the demo is also the trust message for faculty and for the HAM community named as the beneficiary.

4 LITERATURE REVIEW

Strategy used to read the work: inbound content marketing, planned with the RACE journey (Reach, Act, Convert, Engage). Chaffey & Ellis-Chadwick (2019) — owned media and a planned journey outperform one-off promotion when the goal is understanding, not a single transaction. Pulizzi (2014) and Hollebeek & Macky (2019) — content earns trust when it answers the audience's question; engagement follows usefulness, not louder claims. Kotler, Kartajaya & Setiawan (2017) — Marketing 4.0 treats aware-to-advocate as a connected path. Isolated pages do not make a journey.

Gap. Student satellite projects usually stop at a technical report. They rarely publish a public path that both explains the constraint and lets a visitor watch the scheduler decide.

5 METHODOLOGY / WEBSITE DESIGN PLAN

Five pages, one persistent navigation bar. Home also deep-links so the short path and the long path both stay available.



Consideration path: problem and objectives, then how the deployer, spacecraft and ground segment connect, then the KJSIT / KJSSE program map, then the live demo.

RACE	WHAT THE SITE DOES
Reach	Public URL, mission id KJS-SRS-01, headline, and four proof stats: 5 cm, < 1 W, 4 modes, LEO.
Act	Problem cards on Home and “Why manual control does not scale” on Mission.
Convert	Amber “Open ground station” button. Secondary text link: “Read the mission”.
Engage	STREAM LIVE dashboard: phase, SNR, battery, queue, and the reason for the current mode. Commands (force mode, safe mode, SSTV capture) let a visitor participate.

6 IMPLEMENTATION DETAILS

- Page design.** A dark mission-control identity separates the site from a generic college brochure. Headlines stay short, stats are scannable, and one amber action carries the eye. The nav never disappears, so no page is a dead end. Architecture shows SomaiyaPod → SomaiyaSat → ground segment as a block diagram a non-specialist can follow.
- Strategy selected.** Pull marketing through owned content, not paid reach. Audiences on the site itself: students and faculty (KJSIT / KJSSE), collaborator ReOrbit, and the global HAM community as beneficiary.
- Tied to the product.** The router scores each queued item as 0.40-priority + 0.25-urgency + 0.25-link margin + 0.10-power efficiency. Modes below their SNR floor are dropped first. Guard rails run before the policy: no usable link, a 25 s operator override, safe mode below 25% state of charge, and a watchdog if housekeeping is missed. The dashboard prints the rationale — the claim the marketing copy makes, shown rather than asserted.

STUDENTS & FACULTY KJSIT and KJSSE. Recruitment and academic credibility.	COLLABORATOR ReOrbit, Finland, named on the mission site.	HAM COMMUNITY Global amateur radio, the stated beneficiary.
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ROUTER SCORE THE SITE EXPLAINS
0.40 priority + 0.25 urgency + 0.25 link + 0.10 power

1 · Link floor
No usable link, nothing is sent.

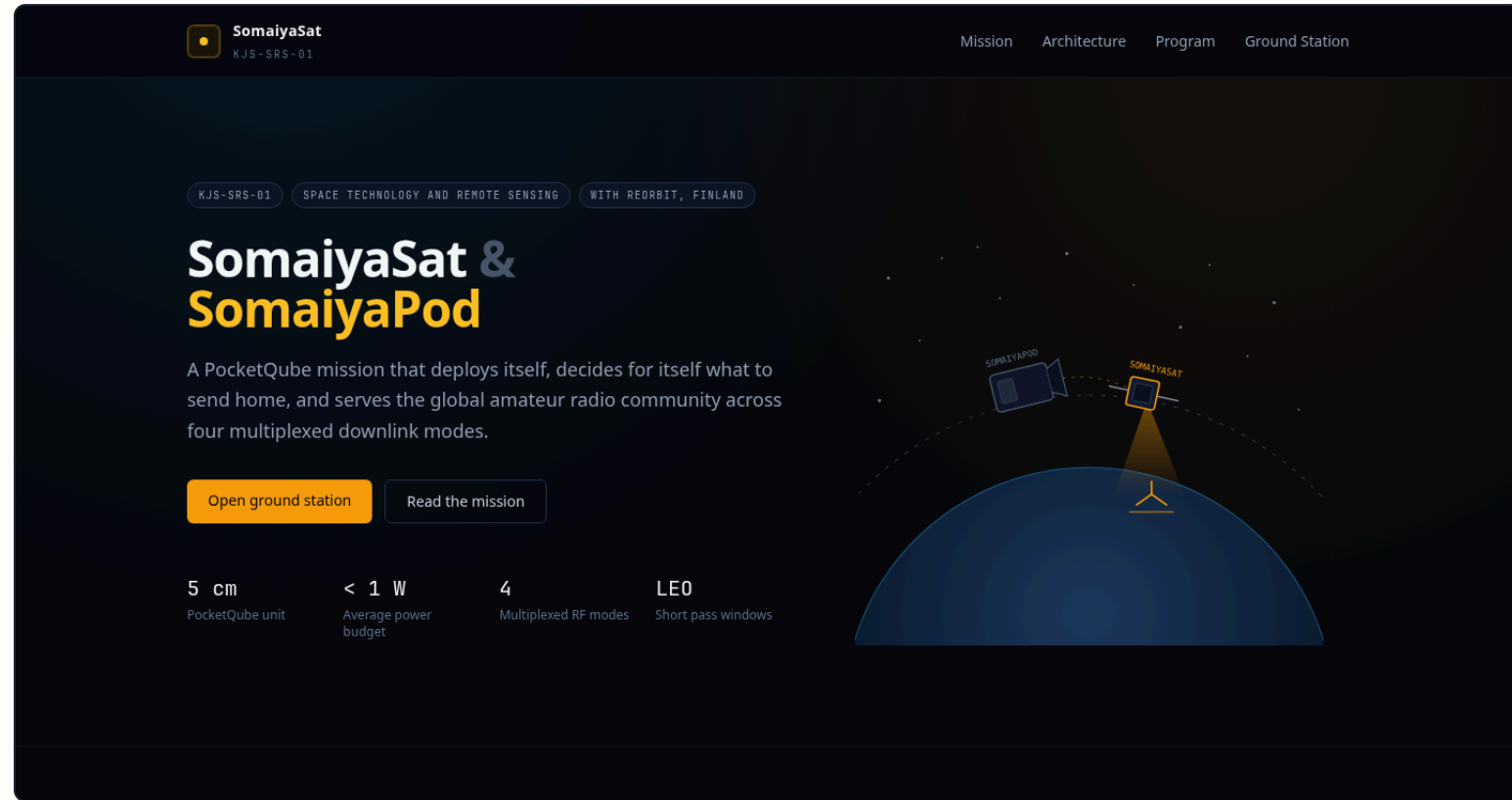
2 · Operator
An uplink wins for 25 seconds.

3 · Safe mode
Under 25% charge, TT&C only.

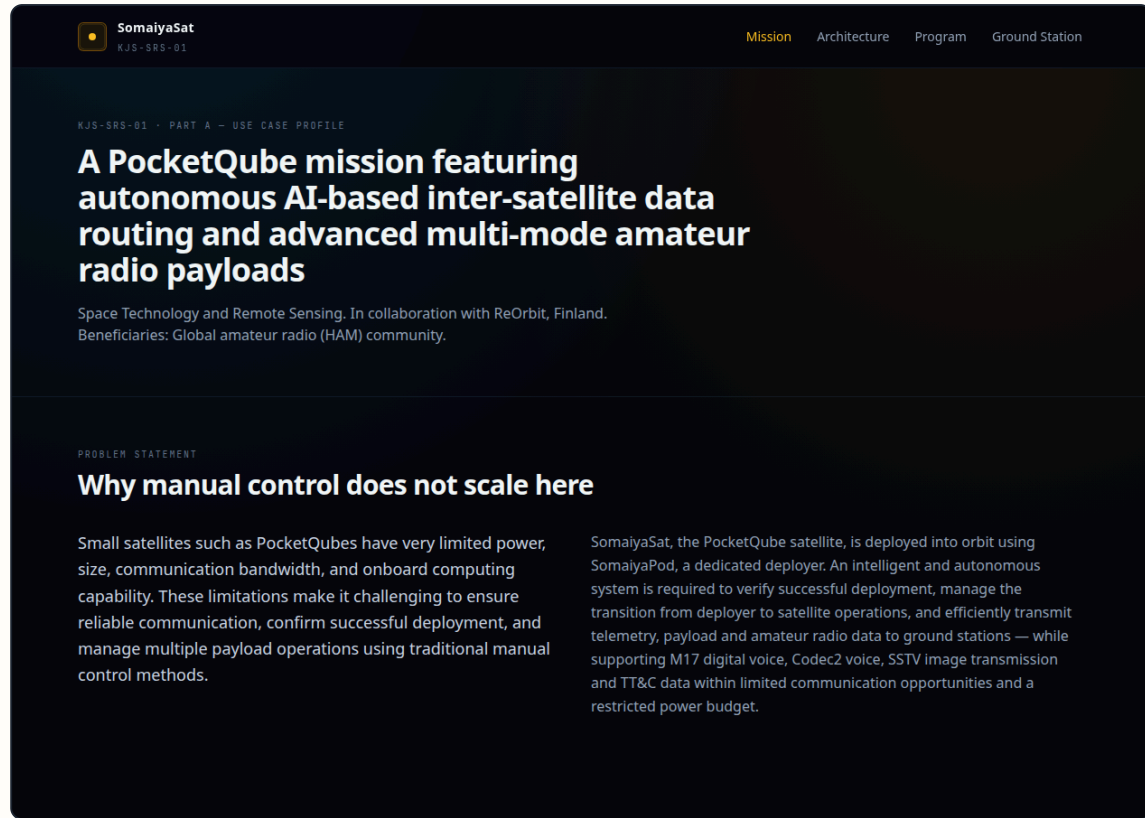
4 · Watchdog
Missed housekeeping forces TT&C.

5 · Policy
Otherwise the highest feasible score.

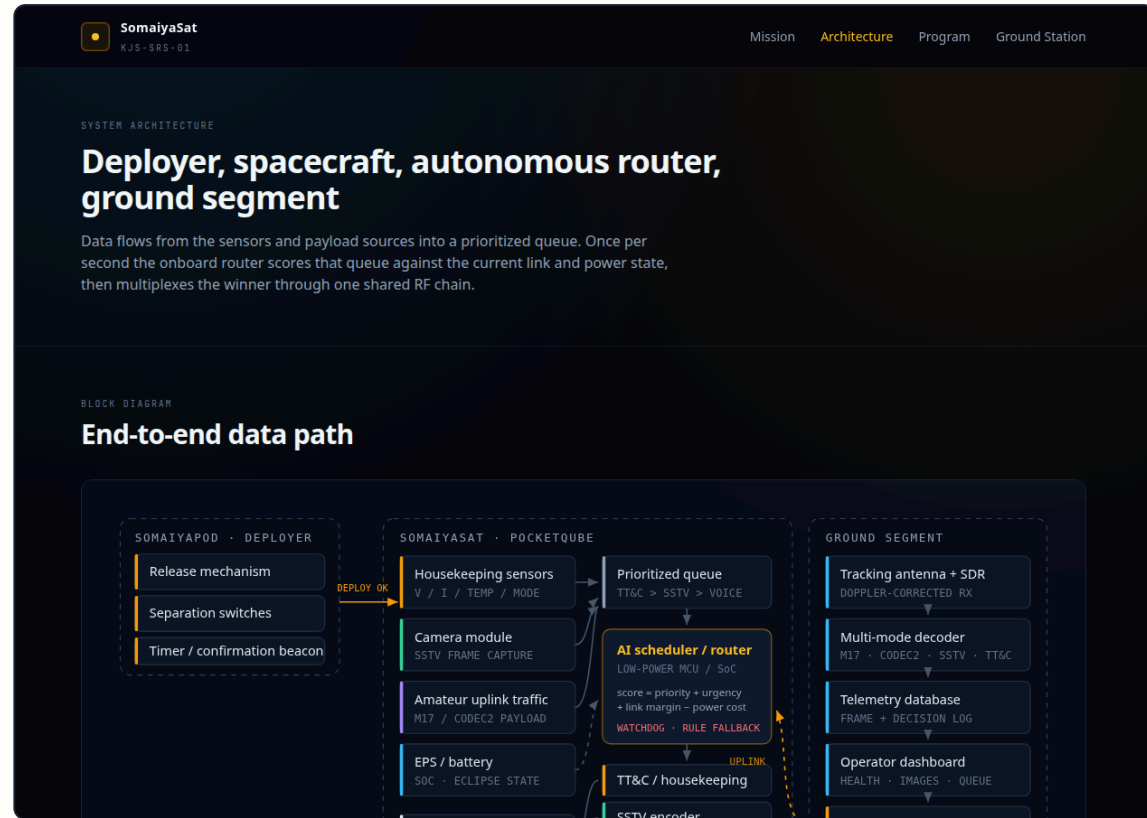
7 RESULTS & DISCUSSION — LIVE PAGES FROM THE DEPLOYED SITE



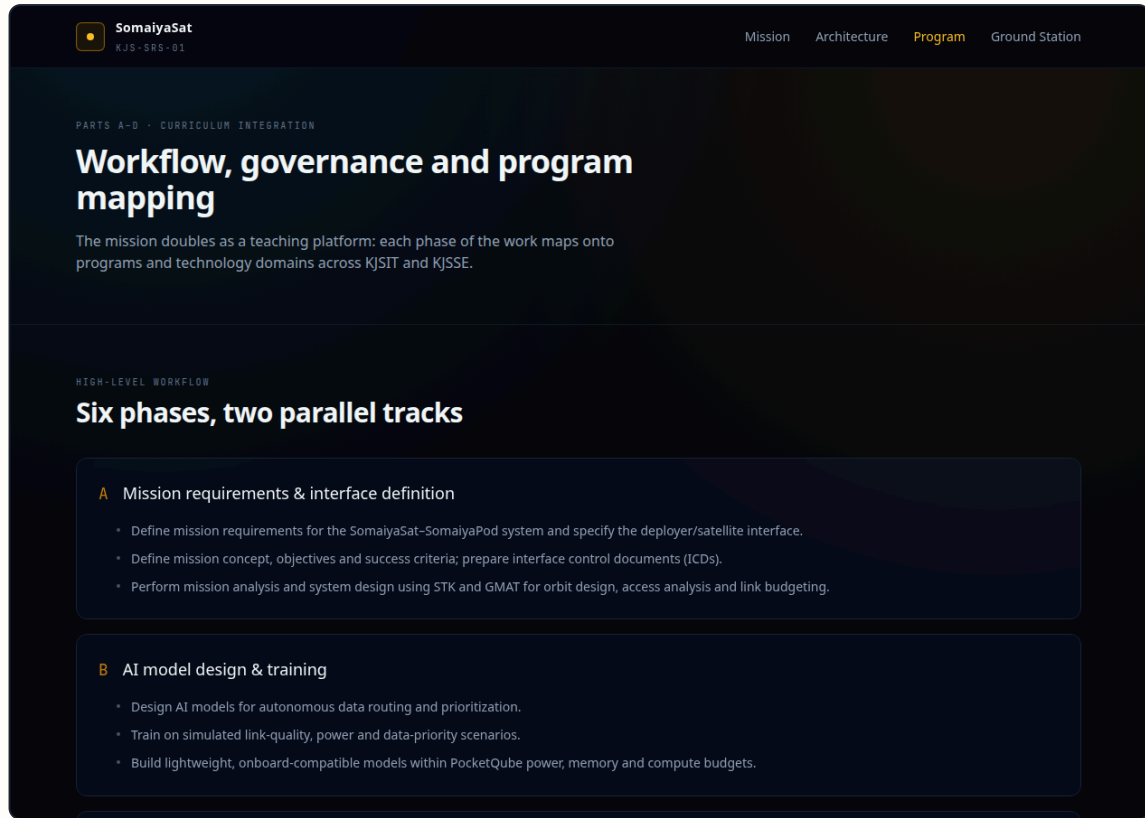
Home. Offer, proof stats, and the amber call to action.



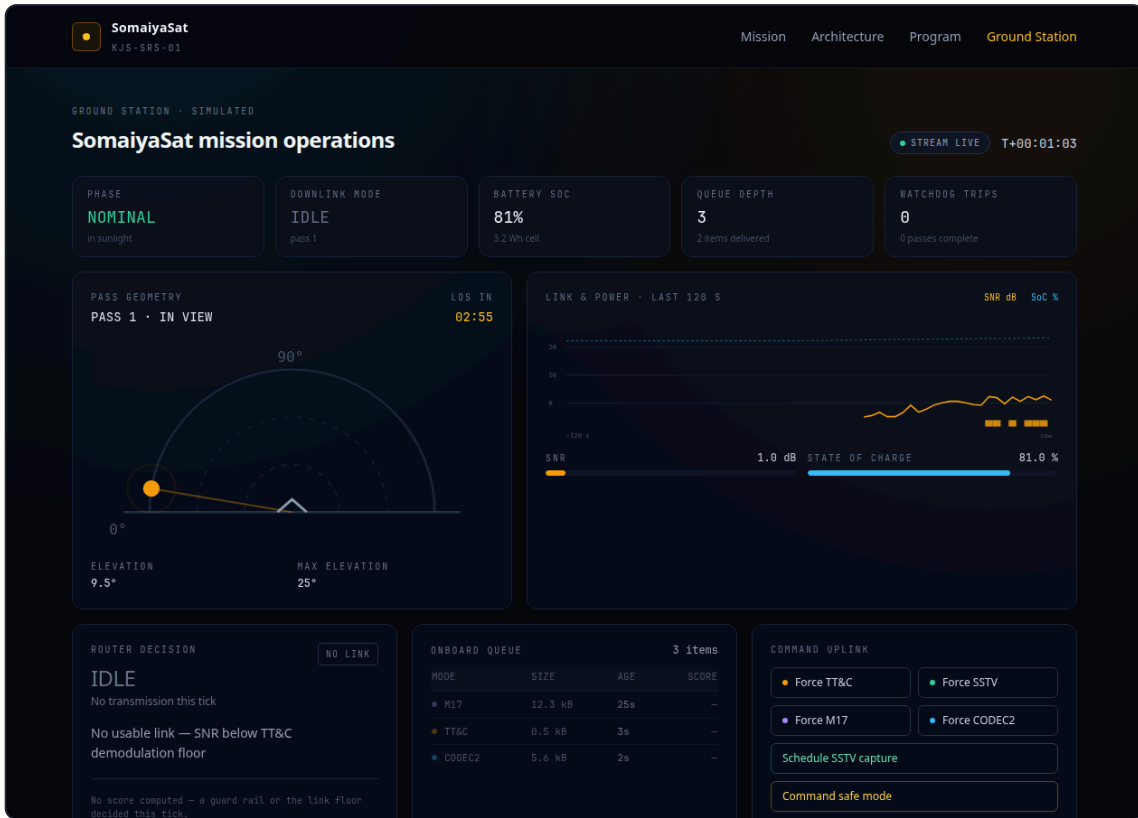
Mission. Use case, beneficiaries, and the problem in plain language.



Architecture. Deployer, spacecraft router, and ground segment.



Program. Workflow and how the mission maps onto KJSIT and KJSSE.



Ground station. Live pass, link, queue, and the router's reason.

8 KEY FINDINGS

- One primary button plus a persistent nav joins five pages. Extra menus would compete with the demonstration.
- The live stream is the engagement asset. Decisions are computed every simulated second, so the ground station can show the link floor, safe mode, and policy.
- Explainable scores turn a power and bandwidth constraint into a trust message for faculty and for the amateur-radio audience.

9 CONCLUSION

The deployed site markets KJS-SRS-01 as a public, explainable mission. A visitor can meet the offer in one click or walk the longer story — problem, architecture, program, then proof. Students, faculty and the HAM community can see both the constraint (minutes of visibility, four modes, sub-1 W) and the response (autonomous routing with guard rails) without a briefing.

10 FUTURE SCOPE

- Measure the journey: page views, CTA clicks, and time on the ground station.
- Extend reach with Somaiya social channels, this poster's QR code, and a one-page brief for amateur-radio clubs.
- Still open: paid reach, a non-compressed link model, and audience testing by batch. The site is a demonstration, not a spacecraft feed.

11 REFERENCES

- [1] Chaffey, D. and Ellis-Chadwick, F. (2019) *Digital Marketing*. 7th edn. Harlow: Pearson.
 - [2] Hollebeek, L.D. and Macky, K. (2019) 'Digital content marketing's role in fostering consumer engagement, trust, and value', *Journal of Interactive Marketing*, 45, pp. 27–41.
 - [3] Kotler, P., Kartajaya, H. and Setiawan, I. (2017) *Marketing 4.0*. Hoboken: Wiley.
 - [4] Pulizzi, J. (2014) *Epic Content Marketing*. New York: McGraw-Hill.
 - [5] Shinde, U. and Nikam, S. KJS-SRS-01 SomaiyaSat & SomaiyaPod. KJSIT / KJSSE. <https://somaiyasat.vercel.app>
- Acknowledgement.** Prof. Poonam Narkhede and Dr. Ankita Nagmothe; use-case owners Dr. Umesh Shinde and Dr. Shailesh Nikam; Department of Information Technology, KJSSE, Somaiya Vidyavihar University.